

AWS Firewall Manager

AWS Firewall Manager is a security management service that enables central configuration and management of firewall rules across all accounts and applications in AWS Organizations.

AWS Firewall Manager simplifies administration of the following protections:

- AWS WAF rules applied to Application Load Balancers, API Gateways, and CloudFront distributions
- AWS Shield Advanced protections for ALB, ELB Classic, CloudFront, Route 53, and EC2 Elastic IPs
- VPC security groups for EC2 instances and Elastic Network Interfaces
- AWS Network Firewall policies deployed across multiple VPCs
- Amazon Route 53 Resolver DNS Firewall rules applied across VPCs

AWS Firewall Manager requirements:

- AWS Organizations must be enabled with all features
- A designated Firewall Manager administrator account must be set in the organization
- AWS Config must be enabled in each member account and region

AWS Firewall Manager key concepts:

- Firewall Manager Policy: specifies which resources to protect and which rules to apply
- Policy scope: defines which accounts and resource types the policy targets
- Automatic remediation: Firewall Manager detects and remediates non-compliant resources automatically
- Common security baseline: new accounts added to the organization automatically receive all policies

AWS Firewall Manager integrates with AWS Security Hub to consolidate security findings. It supports delegated administrator accounts, allowing a dedicated security account to manage all firewall policies without requiring access to the root management account.

Amazon Bedrock

Amazon Bedrock is a fully managed AWS service that provides access to high-performing foundation models from leading AI companies through a unified API, without requiring infrastructure management.

Amazon Bedrock supported model providers:

- Anthropic: Claude 3 Haiku, Claude 3 Sonnet, Claude 3 Opus, Claude 3.5 Sonnet
- Amazon: Titan Text, Titan Embeddings V2, Titan Image Generator
- Meta: Llama 3, Llama 2
- Mistral AI: Mistral 7B, Mixtral 8x7B
- Cohere: Command, Embed
- AI21 Labs: Jurassic-2

Amazon Bedrock key features:

- Model access: each foundation model must be explicitly enabled per AWS region in the Bedrock console
- Amazon Titan Embeddings V2: converts text into 1024-dimensional vectors for semantic search
- Guardrails: content filtering, topic denial, PII redaction applied to model inputs and outputs
- Fine-tuning: customize models using continued pre-training or supervised fine-tuning
- Agents: orchestrate multi-step tasks combining foundation models with external APIs
- Knowledge Bases: managed RAG pipeline connecting foundation models to private data sources
- Prompt caching: reduces latency and token cost for repeated prompt prefixes

Amazon Bedrock is serverless. There are no clusters or instances to provision. Pricing is based on tokens consumed on-demand, or via Provisioned Throughput for reserved capacity. All model invocations are encrypted in transit and at rest. Invocation logs can be sent to Amazon CloudWatch Logs and Amazon S3 for audit and compliance.

Account Factory for Terraform (AFT)

Account Factory for Terraform, known as AFT, is a feature of AWS Control Tower that provides a Terraform-based pipeline for provisioning and customizing AWS accounts at scale.

AFT enables platform engineering teams to define account configurations as code and apply them consistently across all accounts vended through AWS Control Tower.

AFT architecture components:

- AFT Management Account: dedicated AWS account hosting the AFT pipeline infrastructure
- Pipeline infrastructure: AWS CodePipeline, AWS Lambda, Amazon DynamoDB, Amazon S3, AWS Step Functions
- Account requests repository: Terraform configuration files that define which accounts to create
- Account customizations repository: Terraform and shell scripts applied per individual account
- Global customizations repository: configurations applied to every account vended through AFT
- Account provisioning customizations: Lambda-backed pre and post hooks in Step Functions

AFT account vending workflow:

1. Platform engineer commits an account request file to the account requests repository
2. AFT CodePipeline detects the commit and starts the vending workflow
3. AFT invokes AWS Control Tower Account Factory through AWS Service Catalog
4. A new AWS account is created, named, and enrolled in AWS Control Tower
5. AFT applies global customizations to the new account
6. AFT applies account-specific customizations defined for that account

AFT optional feature flags:

- `aft_feature_enterprise_support`: enrolls the account in AWS Enterprise Support
- `aft_feature_aws_config_service`: enables AWS Config recording in the account
- `aft_feature_cloudtrail_data_events`: enables CloudTrail data events logging

AFT prerequisites:

- Terraform version 0.15 or higher
- A dedicated AFT management account separate from the Control Tower management account
- A supported version control system: AWS CodeCommit, GitHub, GitLab, or Bitbucket
- AWS Control Tower must already be deployed in the management account